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Text Visualization Lab Activity. Universal Studios Reviews Analysis Insight Report.

1. Executive Summary

This report answers one question: what turns a Universal Studios visit into a bad experience, and what should each park fix first? It is based on 50,859 TripAdvisor reviews of Universal Studios Florida, Singapore, and Japan, posted from 2002 to May 2021.

Most visitors are happy: 82% of reviews give 4 or 5 stars. The 3,953 negative reviews (1 or 2 stars) follow clear patterns. Long waits are the most common complaint, but they also appear in half of the positive reviews, so visitors mostly accept them. What really separates a bad visit from a good one is **poor staff service** (2.9 times more common in negative reviews) and **rides that are closed or broken** (2.4 times).

Each park has a different main problem:

- **Japan:** queues, crowds, and a language barrier for foreign visitors.
- **Singapore:** rides closed for maintenance, and the heat.
- **Florida:** staff behavior, and visitors comparing it to Disney.

Negative reviews were already rising at all three parks from 2016 to 2019, before COVID. During the COVID period they jumped, reaching 45% of Florida's reviews in 2021, with complaints shifting toward staff and COVID rules.

The main recommendations are to improve staff training (starting in Florida), tell guests about closed rides before they pay, and make the Japan park easier for foreign visitors.

2. Data and Method

2.1 Dataset

Item	Details
Source	Kaggle, "Reviews of Universal Studios" by dwiknrd (originally TripAdvisor)
Documents	50,859 reviews after removing 45 duplicates (about 6 million words)
Branches	Florida (30,587), Singapore (15,743), Japan (4,529)
Time period	October 24, 2002 to May 30, 2021
Non-text attributes	Rating (1 to 5 stars), date, branch

Reviewer names were dropped before analysis, and only short excerpts of review text are quoted. The data is used for academic purposes only.

2.2 Method

1. **Sentiment label from the rating.** 4 to 5 stars = positive (41,680 reviews), 3 stars = neutral (5,226), 1 to 2 stars = negative (3,953).
2. **Text cleaning.** Each review was lowercased, split into words, and cleaned of stop words. Custom stop words included park words ("universal", "park", "day") and place names ("singapore", "osaka"), because they appear in most reviews and say nothing about the

experience. Words were then lemmatized, so "queues" became "queue" and "waited" became "wait".

3. **Text representation.** A document-term matrix (50,859 reviews x 8,489 words) and TF-IDF weights were built to find the most important and most distinctive words.
4. **Complaint themes.** Related words were grouped into 9 themes (for example, "Long waits & queues" = wait, queue, line, hour, minute). A review counts toward a theme if it uses any of its words. Each theme was then compared between negative and positive reviews, across branches, and over time.
5. **Checking with context.** Keyword-in-context (KWIC) lines were read to confirm what the key words actually mean in the reviews.

3. Key Findings

Finding 1: Staff and closed rides drive bad reviews more than waiting does

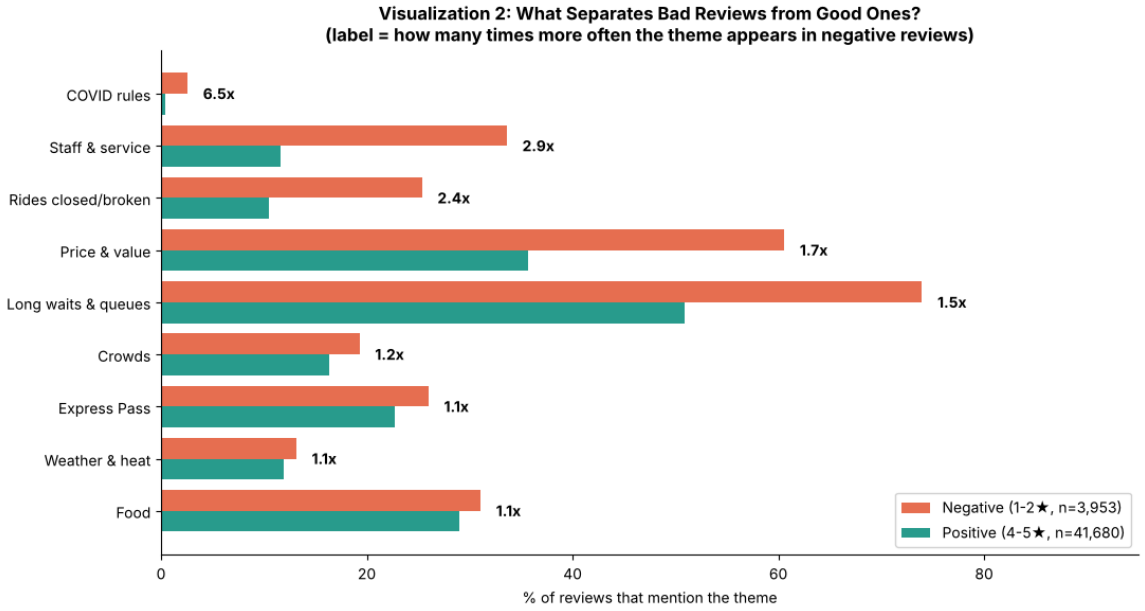


Figure 1. Share of negative and positive reviews that mention each complaint theme. The label is how many times more common the theme is in negative reviews.

Long waits are the most mentioned theme in negative reviews (74.0%). However, they also appear in 50.9% of positive reviews, so the gap is only 1.5 times.

The themes that really separate bad reviews from good ones are:

- **Staff & service:** 33.6% of negative reviews vs 11.6% of positive (2.9 times).
- **Rides closed or broken:** 25.4% vs 10.5% (2.4 times).
- **Price & value:** 60.6% vs 35.7% (1.7 times).

In short, visitors accept waiting as part of a theme park visit. They do not accept rude staff, or paying full price for rides that are shut. Food, weather, crowds, and the Express Pass show almost no difference between negative and positive reviews (1.1 to 1.2 times).

Finding 2: Each park has a different first thing to fix

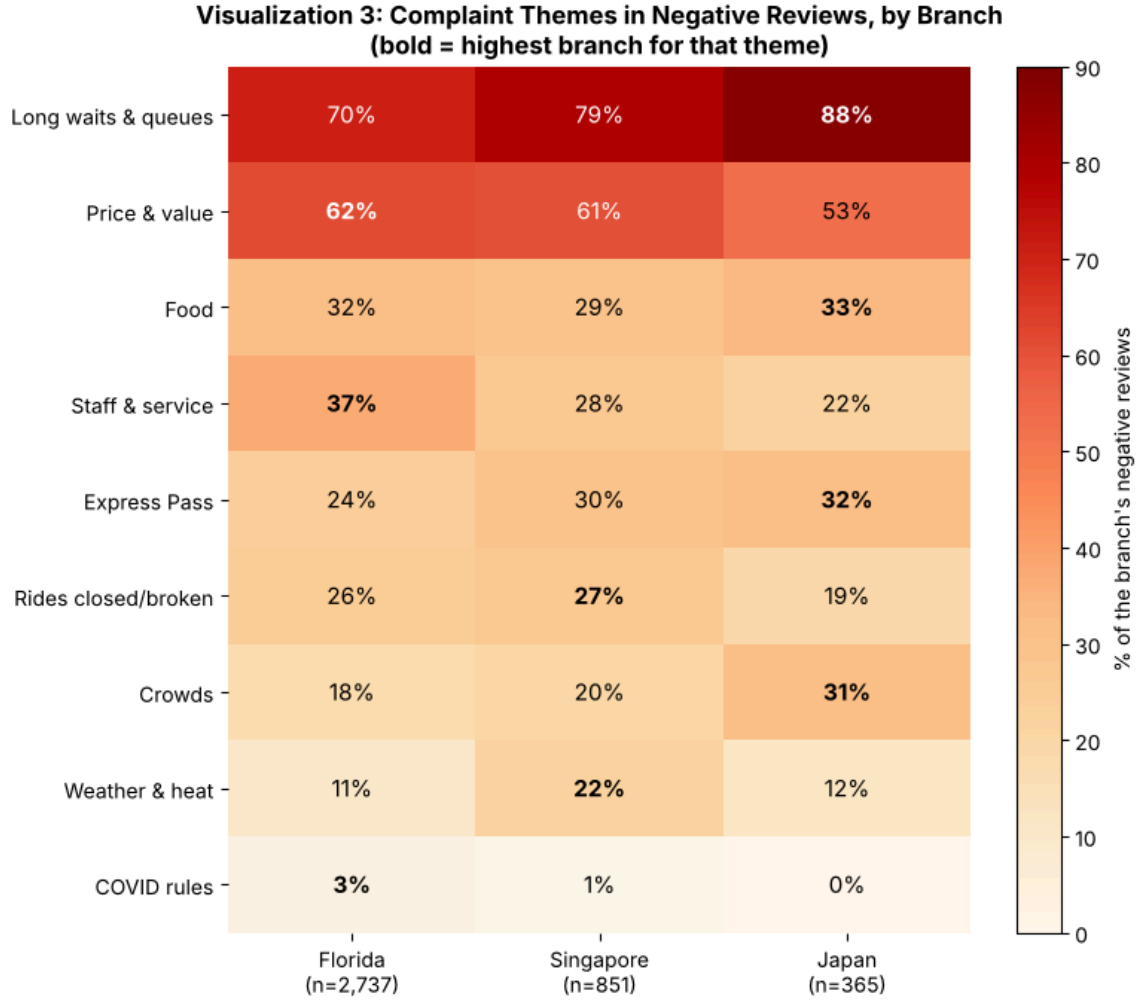


Figure 2. Share of each branch's negative reviews that mention each theme. Bold marks the highest branch for each theme.

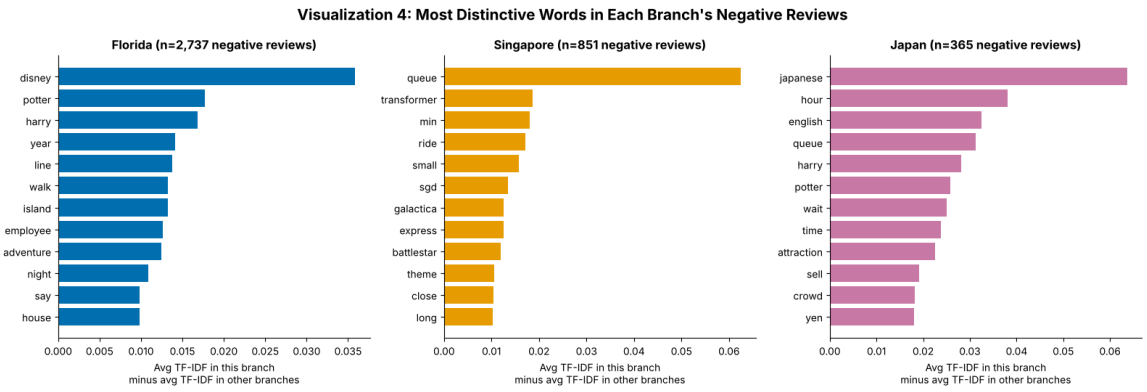


Figure 3. Words with the highest TF-IDF in one branch's negative reviews compared to the other two branches.

Percentages are used because the branches have very different numbers of negative reviews (Florida 2,737, Singapore 851, Japan 365).

1. **Universal Studios Japan: queues, crowds, and language.**
    - Queues appear in 88% of its negative reviews, the highest of the three parks. Crowds (31%) and the Express Pass (32%) are also highest here.
    - "japanese" and "english" are among its most distinctive words. "english" is in 16.4% of negative reviews vs 9.1% of positive ones.
    - The review text shows why: "No English subtitles at all, can't follow story lines" and "the clerk didn't speak any English".
2. **Universal Studios Singapore: closed rides and the heat.**
    - It has the most complaints about rides being closed or broken (27%). "close" is in 20.2% of negative reviews vs 7.6% of positive.
    - Specific rides come up often, such as Transformers, Battlestar Galactica, and Revenge of the Mummy, usually "closed for maintenance" on the day the visitor paid full price.
    - Weather and heat complaints (22%) are twice as common as at the other parks (11 to 12%).
3. **Universal Studios Florida: staff behavior.**
    - It has the most staff complaints (37% of negative reviews, vs 28% in Singapore and 22% in Japan). "employee" is 5 times more common in negative reviews than in positive ones (9.7% vs 1.9%).
    - It also has the most price complaints (62%), and 32.2% of its negative reviews mention Disney, which means unhappy visitors compare it directly to Disney World.

Finding 3: Complaints were rising before COVID, then COVID changed what people complained about

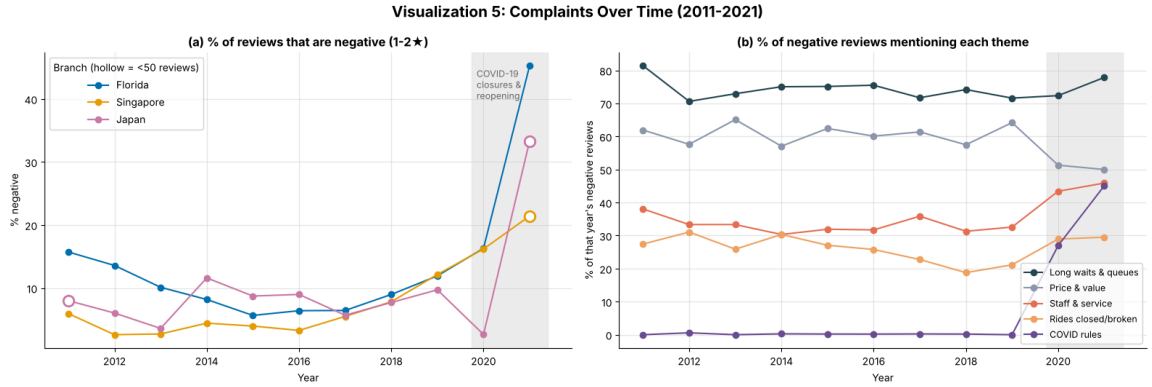


Figure 4. (a) Share of each branch's reviews that are negative, per year. Hollow circles mark years with fewer than 50 reviews. (b) Share of negative reviews that mention each theme, per year.

The share of negative reviews was already going up before the pandemic:

- **Florida:** from 5.7% of reviews in 2015 to 11.9% in 2019.
- **Singapore:** from 3.3% in 2016 to 12.2% in 2019.

In the COVID period it rose sharply, reaching 16.3% for Florida in 2020 and **45.3% in 2021**. The type of complaint also changed. Comparing negative reviews before and after March 2020:

Theme	Before COVID	After March 2020
Staff & service	32.8%	47.3%
COVID rules	0.2%	43.2%
Price & value	61.2%	50.0%
Express Pass	26.6%	15.8%
Long waits & queues	74.0%	73.4%

Many reviews from this period describe arguments with staff over mask and distancing rules, in both directions: some visitors felt rules were enforced too harshly, others felt they were not enforced at all. Queue complaints stayed the same, even with lower park capacity.

4. Recommendations

1. **Fix the staff experience first, starting with Florida.**
    - Train staff on how to talk to guests, and on applying rules (lines, ride rules, health rules) politely and the same way every time.
    - Put more staff on crowd control and directions at the busiest rides and entrances.
    - Target: bring the share of Florida negative reviews that mention staff from 37% down to the 22 to 28% of the other parks within one year.
2. **Tell guests about closed rides before they pay.**
    - List closed and under-maintenance rides on the ticket website and at the gate, for all three parks.
    - Offer a discount or a free return visit when a major ride is closed on the day of the visit.
    - Singapore should schedule maintenance in the low season and add shade, fans, and water stations in outdoor queues.
    - Target: lower the share of Singapore negative reviews about closed rides (27%) to below 20%.
3. **Make Universal Studios Japan easier for foreign visitors and manage its crowds.**
    - Add English subtitles for shows, or a subtitle app guests can use on their phones.
    - Train staff at ticket counters and Express Pass sales in basic English.
    - Show live wait times and crowd forecasts in English, so tourists can avoid peak days.
4. **Keep monitoring reviews with this method.**
    - Negative reviews were rising even before COVID, so each park should repeat this theme analysis every quarter.
    - A rise in staff or closed-ride themes should trigger a review by park management, since these are the strongest signs of a bad visit.

5. Limitations

1. **Self-selected reviewers.** People with very good or very bad visits are more likely to post on TripAdvisor, so the reviews do not represent all visitors.
2. **English only.** The Japan data mostly reflects foreign tourists, not local visitors. This partly explains why language is a top complaint there.
3. **Keyword-based themes.** The themes count words, not meaning. Removing stop words also removes "not", so "not crowded" counts as a crowd mention. This is why themes are compared between negative and positive reviews, where the same error happens in both groups.
4. **Small COVID sample.** The data ends in May 2021. After March 2020 there are only 189 negative reviews from Florida, 31 from Singapore, and 2 from Japan, so the COVID findings mainly describe Florida.
5. **Old data.** The newest review is from 2021. The parks may have already fixed some of these issues, so the findings should be checked against newer reviews before acting on them.